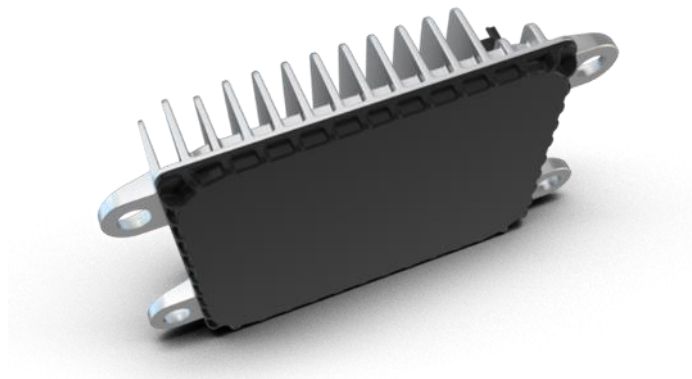


Product Data Sheet

ProWave Gen1 - Common

ZF Friedrichshafen AG



Version 1.39

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Abbreviations and Definitions

Abbreviations

AD	Autonomous Driving
ADAS	Advance Driver Assistance Systems
AppPar	Application parameter
CFAR	Constant False Alarm Rate
dBsm	Decibel Per Square Meter
FCC	Federal Communications Commission
ETSI	European Telecommunications Standards Institute
IMU	Inertial Measurement Unit
LLDP	Link Layer Discovery Protocol
FMCW	Frequency Modulated Continuous Wave
FoV	Field of View
FRR	Full Range Radar
GM	Grand Master
gPTP	Generalized Precision Time Protocol
HSM	Hardware Security Module
MIMO	Multiple Input Multiple Output
MSE	Mean Square Error
ODD	Operational Design Domain
P2P	Peak to Point
PRT	Pulse Repetition Time
PTL	Point Target List
RCS	Radar Cross Section
RF	Radio Frequency
RG	Range Gate
SEooC	Safety Element out of Context
STMF	Single Target Model Fitting
TTMF	Two Target Model Fitting
UDP	User Datagram Protocol

Definitions

Resolution

Resolution is the ability to separate two equally strong targets in one dimension if the other dimensions are the same.

Accuracy

Accuracy is the degree of agreement between the measured dimension for a target (e.g., range) and the true value of the dimension for this target.

Precision

Precision is defined as the degree of agreement between a set of values of a measured target property (e.g., range) obtained from multiple repeated measurements of the identical object under equivalent conditions.

Ambiguity

Ambiguities are present in the two angular dimensions and in the velocity dimension. Ambiguities are encountered in a dimension if the frequency is sampled below the Nyquist criterion. For example, a maximum unambiguous velocity of 53 m/s characterizes the radar. If the relative velocity of a target to the ego-vehicle is 60 m/s, the detected velocity from the radar will be ambiguous, i.e., $60 - 53 \text{ m/s} = 7 \text{ m/s}$.

The ProWave Gen1 has implemented certain features to solve the ambiguities.

FoV

The FoV is the image section in which the radar sensor is able to detect objects with a certain resolution and accuracy.

Peak

A peak is an initial version of a final point target. The differences between a peak and a point target are:

- a. Angular ambiguities are not solved yet
- b. Each dimension is represented by the dominant FFT bin and is still not interpolated
- c. Peaks can consist of merged reflections from more than one scatterer spaced less than one FFT bin apart

Point-Target

A point target represents a single radar detection.

PRT

Pulse repetition time, which describes the grid for single frequency chirp (pulse).

CFAR

Constant false alarm rate is a method achieved by adapting the detection threshold on the noise in the close environment of the individual detection.

AppPar

AppPar are application specific parameters, which can be used to tune the radar sensor to the needs of the specific application.

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1 Document Scope

This document is a product datasheet for the ProWave radar sensor. It is a first level information about the measurement capabilities and the features of ProWave radar. All the information described in the data sheet is correct. However, it could be that some detailed information of a feature is not mentioned in the data sheet.

2 Product Description

The ProWave is a high-resolution full range automotive radar sensor, which is operating in the 77GHz band only. The sensor can accurately identify the 4D location of a target (range, relative speed, azimuth, and elevation).

Due to the very high resolution in Doppler dimension (a delta of 0.05m/s in relative speed can be separated) the sensor offers a great overall detection performance which gets close to Lidar resolution capabilities.

In contrast to common radar sensors especially for well-known AD-functions, like e.g., ACC, the ProWave is designed as a pure point cloud sensor and outputs just detection points in polar coordinates without any transition to the vehicle coordinate system. There is no mounting window defined since the sensor does not offer any function except for the detection of reflection points. The interpretation of this information must be done on the host side.

Especially the sensor does **not** include:

- Misalignment detection or compensation
- Blockage detection
- Clustering or Tracking
- Free space calculation

The sensor is designed as Safety Element out of Context (SEooC) for ASIL B (D) applications and meets three safety goals

1. There are no false positives caused by EE failure
2. There are no false negatives caused by EE failure
3. The sensor detects internal EE failures, which could cause false positives or negatives, within a defined time

The Radar sensor offers some flexibility in operation, which can be parameterized:

- Dynamically by a COM message
- Statically by application parameters (AppPar) stored in the non-volatile memory

3 Field of Application

3.1 Assumption of Use

The sensor is designed for AD-applications, where multiple sensors are connected to a central ECU that does the post processing of the radar detections. Since such vehicle systems typically have a complex power supply to serve all different ECUs with an appropriate supply voltage, the sensor is offered with a 12V input power supply rail only.

Since the sensor is designed as Safety Element out of Context (SEooC), it is assumed that significant safety measures are implemented on the host system, which runs the perception algorithm and the strategy of the driving function.

The sensor cannot guarantee that all the delivered targets are correctly placed (false positive) or that no target exists where the sensor reports no target (false negative). The sensor itself runs a couple of self-diagnostic monitors to ensure that these kinds of errors never occur due to an EE-failure. However, there are physical effects causing false positives or false negatives, which cannot be avoided, such as:

- Mirroring of radar beams on the road surface
⇒ *This would lead to detections beneath the road surface*
- Mirroring of radar beams on lateral objects, e.g., a guardrail
⇒ *This would lead to detections outside of the road*
- Reduction of SNR in the environment of another (strong) target
⇒ *This would lead to non-reported small detections like a pedestrian next to a truck*

3.2 Safety Goals

The ProWave sensor is developed as a SEooC. The sensor is compliant to the functional safety standard ISO 26262 with the safety requirements listed in Table 1:

Table 1: Safety requirements on sensor level

	Safety requirements on sensor level	ASIL	Comment
SG01	E/E related faults which have the potential to cause loss of Detection of point targets shall be detected	B(D)	False negatives: it cannot be guaranteed that all targets will be detected by the radar due to technological challenges. However, loss of detection due to internal E/E failures is rated ASIL B(D) .
SG02	E/E related faults which have the potential to cause Misdetection of point targets shall be detected	B(D)	False positive detections. It cannot be guaranteed that all point targets result of a real present target due to e.g. mirroring effects. However, loss of detection due to internal E/E failures is rated ASIL B(D) .
SG03	The system shall identify internal failures which could create a false negative or false positive within a certain time	B(D)	The system can guarantee that all internal failures are detected within 360ms.

4 Sensor Description and Cycle Timing

4.1 General Description of Operation Principle

The ProWave is a monostatic automotive radar sensor, which operates with 12 TX and 16 RX channels of a cascaded MMIC array with 4 single chips. The sensor operates all RF ports in parallel and therefore achieves $12 \times 16 = 192$ synthetic MIMO channels. 128 channels are used for digital beamforming and the other channels are used for angular ambiguity solving.

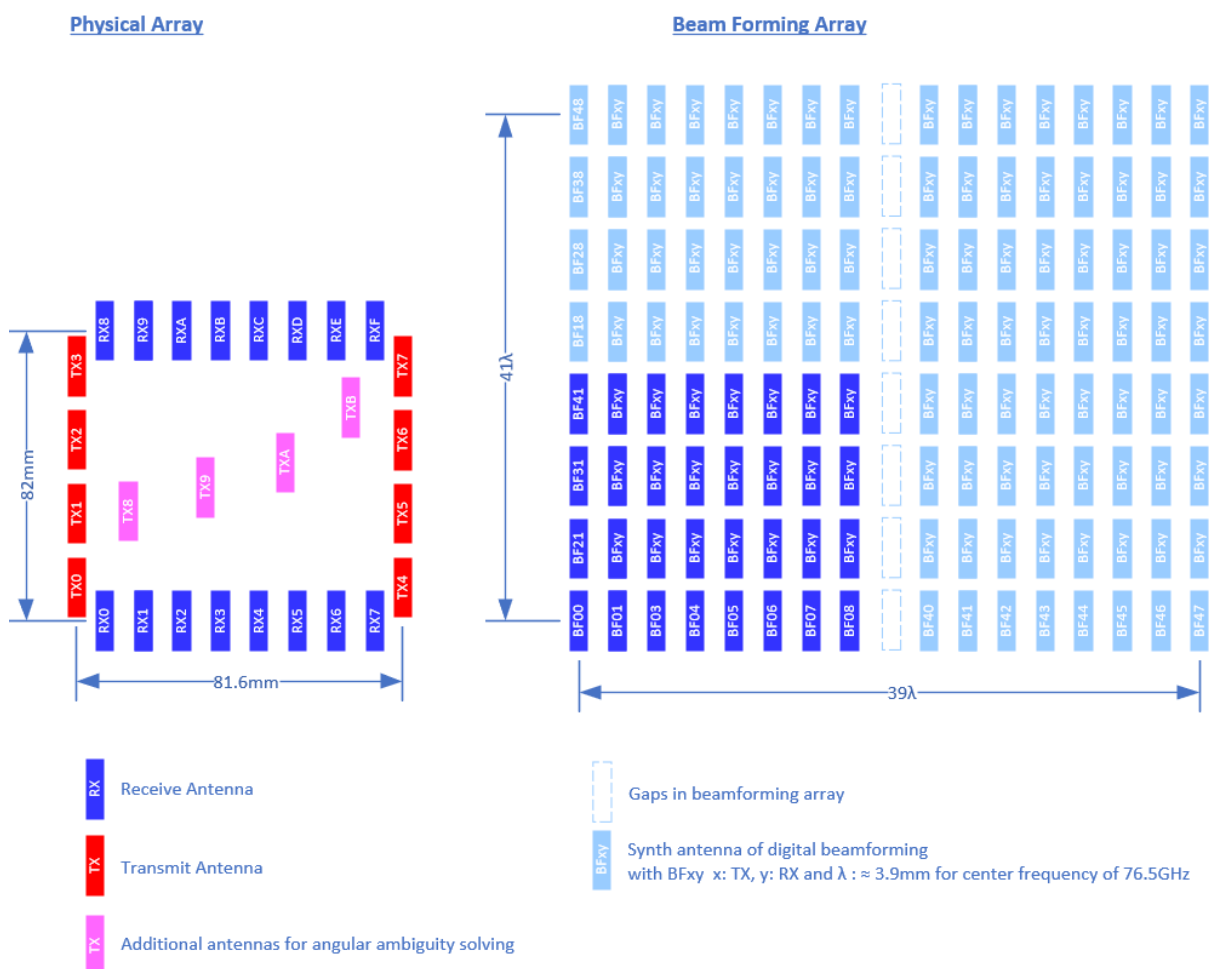


Figure 1: Antenna Array

Furthermore, the sensor employs a chirp-sequence modulation scheme with 512 samples per chirp and 512 chirps in one radar cycle. The data acquisition phase alternates between 19 and 25ms, due to two different PRTs. This feature enables the receiver of the point cloud to solve the speed ambiguity by observing the velocity of a single target over two consecutive radar cycles.

The bandwidth of the chirps can be controlled either by COM or by application parameter. There are 3 predefined sweep bandwidths implemented that allow the application to adjust the range resolution to the needs of the scene interpretation. Since the ProWave always calculates 512 range gates, the maximum unambiguous range is also impacted by the selected

range resolution. The effective bandwidth does not impact the general sensitivity of the radar system, which can detect a 10 dBsm target up to 350m but is limited by the SW.

The main waveform of the transmitted radar signal can be seen in figure 2.

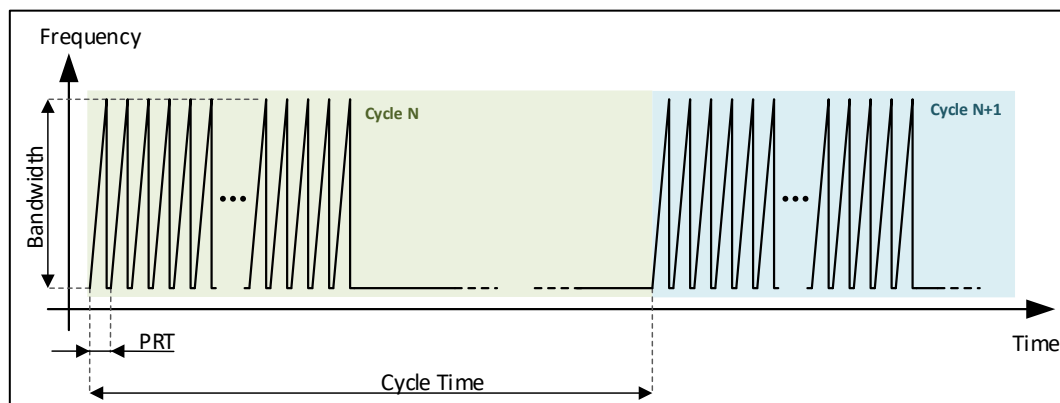


Figure 2: Operation principle using frequency modulated signals and fast chirps

The radar operates in the 76-77 GHz frequency band. As mentioned previously, the radar uses three different sweep bandwidths to generate different range resolutions. The effective bandwidth that is employed during ADC sampling is always a bit smaller than the chirp sweep since the synthesizer in the RF-chip needs some time to settle before the linearity of the chirp is guaranteed.

4.2 Cycle Timing

Figure 3 shows the internal timing within the radar sensor. Due to parallel processing the ProWave can achieve both, a fixed radar cycle time, and a low latency from scene change (center of data acquisition) to transmission of PTL.

The radar cycle time is a fix value (min 60ms or adjusted; see application parameter) and it is not scene dependent.

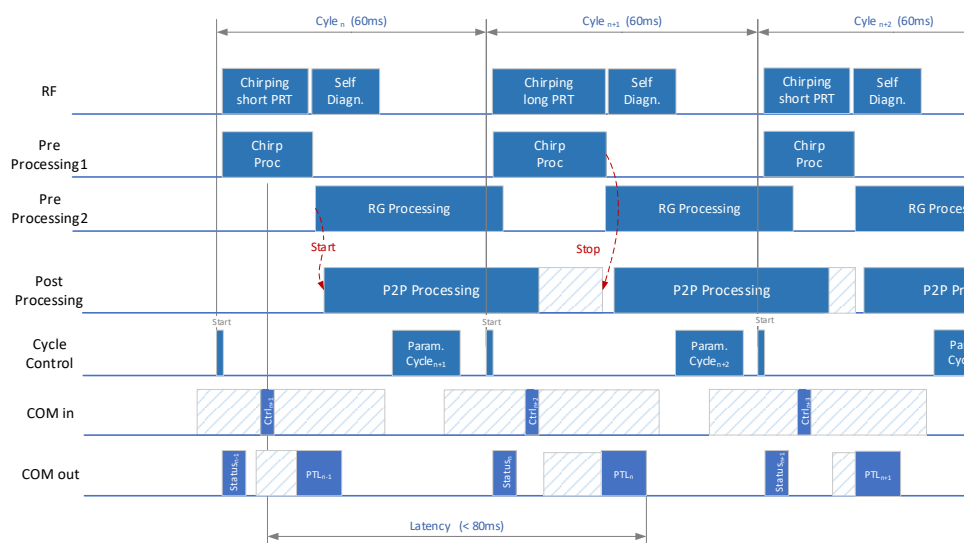


Figure 3: Timing Diagram

4.3 Radar Signal Processing

The radar signal processing is divided into 3 stages (see Figure 3):

- Chirp processing
- Range gate (RG) processing
- Peak-to-point (P2P) processing

Chirp processing is doing mainly the range FFT, but also interference filtering in the time domain.

RG processing is doing

- Doppler FFT
- MIMO decoding
- Angular FFTs (AZ + EL)
- CFAR detection
- Peak extraction

P2P processing, which is done during post processing of the radar peaks, does:

- Calculates interpolated frequency & power values between the FFT-bins for all dimensions
- The split of one single peak in two-point targets by a two-target model fitting, if possible
- Selects the most likely unambiguous angle in the azimuth and elevation dimension
- Calculates the MSE values and confidence value of each single point target

If the two-target model fitting can be applied, the resolution numbers in brackets are achievable (see Table 2).

4.4 Time Synchronization with gPTP

The ProWave supports high-precision time synchronization based on IEEE 802.1AS, also known as gPTP. Due to automotive requirements the implemented time synchronization over Ethernet uses the current mechanisms as defined in IEEE 802.1AS with specific extensions and/or restrictions, since Ethernet components inside a car are static and their role and function is known.

The gPTP integrated in ProWave uses the exchange of time-stamped messages to communicate time from a grand master clock to several bridge and end-point devices. Regarding to IEEE 802.1AS the ProWave expects a 2-step peer-to-peer synchronization between his neighboring time-aware system where the ProWave is acting as time slave.

The following time-stamped messages are supported (see Figure 4):

- Sync,
- Follow_Up,
- Pdelay_Req,
- Pdelay_Resp and
- Pdelay_Resp_Follow_Up

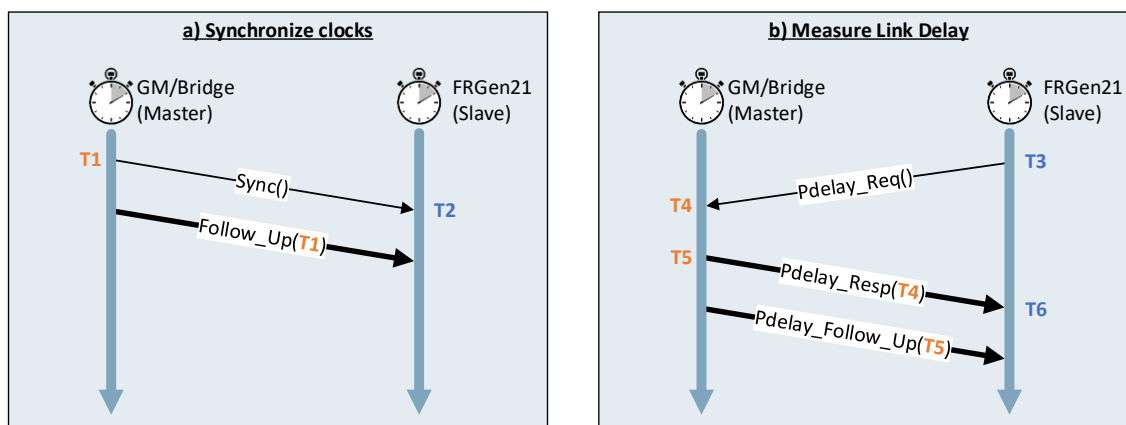


Figure 4: gPTP implementation in ProWave

The clocks between master and slave (ProWave) are synchronized by sending the Sync and Follow_Up message. The link delay is measured in a separate sequence with Pdelay_Req, Pdelay_Resp and Pdelay_Follow_Up. Both sequences are independent but need to be done to calculate the slave clock offset.

From startup, the ProWave transmits Pdelay_Req messages to the connected master. These cyclic messages are sent via LLDP multicast. The ProWave does not support announce messages to support the best GM selection algorithm. In the case of an automotive application, the role of the involved components for gPTP synchronization should be preconfigured. The ProWave acts as a time-aware end-station with a slave port.

If the time synchronization with a master is successful, the ProWave adjusts its internal time base as the time difference in nanoseconds since Jan 1st, 1970 0:00am.

If the time synchronization with a master is not successful, the ProWave adjusts its internal time base as the time difference in nanoseconds since power up.

The status of the time synchronization can be derived from the variable "TimestampMeasureStart" in the header of the PTL. Values greater than 1.5E18 nanoseconds indicating a valid synchronization, while much lower values (lower than e.g. 8.7E13 nanoseconds == 1 day) indicate an asynchronous operation.

The accuracy of the time synchronization achieves approximately $\pm 500\mu\text{s}$. The ProWave guarantees a precision of the start of radar cycle timing of $\pm 500\mu\text{s}$ referred to as synchronized time base.

4.5 Key Parameters of Radar Detection

The detection performance of the ProWave is shown in table 2. All values are defined for a single target with a defined RCS of 10 dBsm.

Table 2: Definition of Radar Detection Performance for all available modes

Parameter		Unit	Short	Mid	Long	SFMCW-M ^{*6)}	SFMCW-L ^{*6)}
Range	FoV ^{*3)}	[m]	100	200	350	260	350
	Resolution (HR)	[m]	≤ 0.21 (≈0.1)	≤ 0.41 (≈0.2)	≤ 0.77 (≈0.4)	≤ 0.21 (≈0.1)	
	Precision ^{*5)}	[m]	≤ 0.02	≤ 0.04	≤ 0.07	≤ 0.02	
Speed	FoV	[m/s]	alternating cycle by cycle 0...38 or 0...53				
	Resolution (HR)	[m/s]	0.1 (≈0.05)				
	Accuracy	[m/s]	0.01				
Azimuth	FoV (guaranteed range ^{*4)})	[°]	0 ... ± 20 (300m) ± (20 ... 45) (150m) ± (45 ... 60) (80m) ± (60 ... 80) (detection yes, no guaranteed range)				
	Resolution	[°]	1.5 @ +/-20° (0.8 @ +/-20°)				
	Accuracy ^{*2)}	[°]	0.15 @ ± 20° 0.50 @ ± (20° ... 45°) 1.00 @ ± (45° ... 60°)				
Elevation	FoV (guaranteed range)	[°]	± 12 (250m) ± 15 (175m)				
	Resolution	[°]	1.5 @ ± 15°				
	Accuracy ^{*2)}	[°]	0.15° @ ± 15°				
Auxiliary	Radar cycle time	[ms]	60 ^{*1)}				
	Frequency bandwidth	[MHz]	860	450	260	950	
	Average power consumption	[W]	typical 18 / worst case 23				
	Peak power consumption	[W]	≈31 (during 19ms chirping)				

^{*1)} it can be adjusted to higher values by AppPar.

^{*2)} angular accuracy is given under the assumption that the far field condition is fulfilled (range > 3.4m).
Below far field limit, detection is guaranteed but angular accuracy can't be achieved.

^{*3)} max range is defined by the number of range gates (512) times the range gate length – $\approx 5\%$

^{*4)} attenuation of secondary surface (radome) is **not** included

^{*5)} range precision is defined by the standard deviation of the range coordinate over a scan of the azimuth FoV

^{*6)} SFMCW modes **not** validated for serial use

There are some additional comments mentioned below to the different measurement values.

Range

The FoV in range is defined by the selected chirp bandwidth (see description in chapter 4). The host can select one of the 3 predefined effective bandwidths but not define a complete new one. Targets at a distance greater than the maximum unambiguous range (depending on the chirp bandwidth) will be suppressed by the radar and will be not visible in the PTL.

Speed

The unambiguous speed measurement range is limited but can be extended by observing the speed values of a point target over two consecutive cycles. Since the PRT is alternating from cycle to cycle, the speed measurement of a target outside the unambiguous FoV varies from cycle to cycle as well. One can solve the unambiguities by applying the Chinese remainder theorem (see Figure 5). This algorithm is currently not running on the radar sensor.

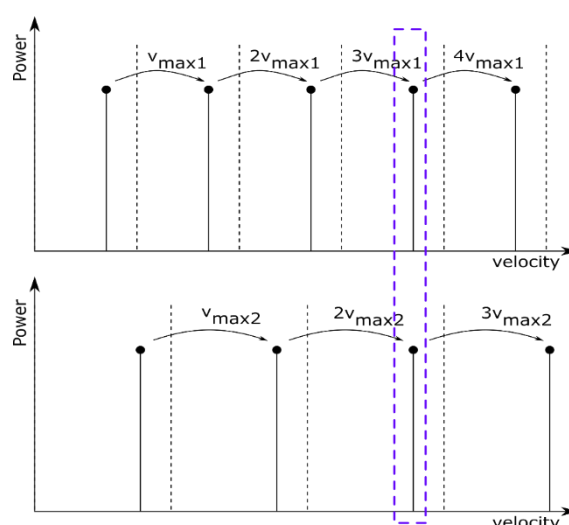


Figure 5: Chinese remainder theorem

Angular FoV

Detections outside of the angular FoV are not suppressed and will be reported as point targets. However, such detections are possible for short ranges only since the sensitivity of the antennas is limited (see diagram below).

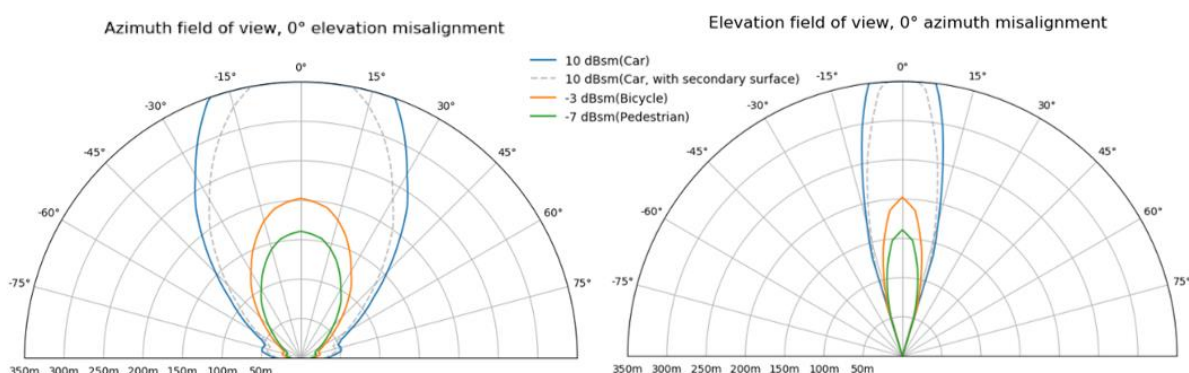


Figure 6: FoV related to the antenna pattern for targets of different reflectivity

No misalignment compensation has been considered for the FoV in angular dimension.

4.6 General Comments to Detection Probability

The ProWave is a pure point cloud sensor, which reports what it detects in its FoV. The sensor does not include misalignment compensation because it reports its detection in polar coordinates (sensor-centered coordinate system).

The detection of a target is not guaranteed by 100% even if the radar sensor is working well. The reasons for such phenomena can be:

- Reduced visibility or even invisibility
- Detection gaps caused by the destructive interference of the direct and indirect paths the reflection follows on its way to the sensor. The indirect path can be caused by mirroring on road surface, guardrail, etc.
- Interference by other radar sources which increases the noise level
- Blockage due to snow which reduces the signal strength
- Side lobes of the applied signal window which leads to a limited detection close to stronger targets
- Artifacts in Doppler dimension due to TX MIMO operation
- Ambiguities in the angular dimensions
- Exceeding of computational time budget

- Mirror effects
 - The same effect can lead to point targets with an elevation height ($\text{range} * \tan(\Theta)$) which is below the road surface
 - Targets can be placed outside of the road due to mirror effects on guardrails
 - Ghost targets can be created by multipath reflections

Since the radar typically operates in a dynamic environment, most of these effects happen just in one or a few cycles (e.g., artifacts from TX MIMO operation, angular ambiguities).

Mirroring effects can be handled by a perception system/algorithm. Such a system is able to identify where the road surface is and where the guardrails are. Hence, mirrored detections can be handled either by correcting the mirrored dimension or by filtering them out of the PTL.

Blockage and interference can be solved by observing certain parameters using history information (available in a perception algorithm). The sensor itself indicates the presence of interference (if detected), but a blockage detection algorithm running outside of the sensor shall quantitatively assess performance degradation.

4.7 Sensor Support for Detection Probability Estimation

The ProWave delivers some parameters per each target point in the PTL, which can be used to estimate the detection probability. These parameters are listed below.

MSE_SNR	Offset: 0x08
MSE_Range	<i>It indicates the model fitting error in each dimension.</i>
MSE_Velocity	
MSE_Azimuth	
MSE_Elevation	
MSE_SubArray	<i>It indicates the agreement of the sub-array steering vector with the respective ideal model. A similar value of both MSE Subarray and MSE Subarray 2nd best indicates low confidence in ambiguity solving.</i>
MSE_SubArray 2nd Best	
SNR	<i>Ratio of point target power to CFAR noise</i>
ModelType_Confidence	Offset: 0x16
ModelType	<i>It indicates whether one or two-target model has been applied but there are more options (please see documentation to PTL)</i>
DetectionConfidence	<i>It is weighted sum of all the MSEs (the combination is IP of ZF). Note: It is up to a customer to define its own weighted confidence based on the MSEs (if needed).</i>

4.8 Parameterization

4.8.1 General

The radar sensor operation can be basically tuned in two ways

- Via COM messages
- Via application parameter (AppPar)

Tuning via COM message is possible for

- Range mode (Chirp bandwidth = range resolution)
- PRT mode

Tuning via AppPar is possible for

- COM parameter (IP/port addresses, net mask, etc.)
- Range mode control and range mode sequence
- PRT mode control and PRT sequence
- Cycle time and time synchronization
- Interference mitigation measures (frequency multiplexing, up/down chirp)
- Thresholds tuning

All modes are described in detail in the following sections.

4.8.2 COM Parameter

The sensor has one single Ethernet channel. It is a 100BaseT BroadR Reach physical layer.

The 2nd Ethernet channel is reserved for later usage and is currently not populated.

The MAC address of each sensor can be read from the sensor, but it cannot be changed. ZF guarantees that the MAC address is unique for every sensor.

On top of this one single HW channel, there are multiple SW channels available.

The following parameters can be changed individually

- IP addresses of sensor and host
- Port addresses of sensor and host for the PTL and status data

The communication elements are assigned to the different channels as follows

1. Main COM port
 - a. PTL output
 - b. Input data input
 - c. gPTP input and output
 - d. UDS protocol input and output
2. Second COM port
 - a. Status data output

The following table shows the default settings.

Table 3: Default IP configuration

Name	Value
Sensor IP address	192.168.100.60
Host IP address	192.168.100.30
Main port	60001
Logging port	60000

4.8.3 Range Mode Selection

The range mode selection gives the application system the choice of whether the radar shall detect with the best available range resolution or up to far distances. The radar itself always operates with the same antenna but changes the range resolution by adapting the chirp bandwidth. Since the radar is just able to process 512 range gates, adapting the range resolution has an impact on the maximum range as well. The maximum range is defined by range gate length (FFT bin grid) times 512. Since the noise for the last 8-10% of the range gates is slightly raised, the given maximum range of the far range mode is a bit shorter. This effect is valid for all modes.

The range mode can be controlled either via AppPar or via COM, selected by the parameter "RMP mode".

4.8.3.1 AppPar control

If the parameter "RMP mode" is set to AppPar, the range mode is defined by a sequence called "RMP sequence".

Figure 7 shows the parameter field for the range mode with an iterative sequence over all elements. In such case, all elements are defined by a range parameter and the sequence wraps to element "0" after element "8". However, the sequence can be shortened by defining an element with the "end of sequence ID". Now the sequence wraps around to element "0" if the following element contains an end of sequence ID (end of sequence ID: 0xFF).

After power up, the Radar always wakes up with the range mode defined in element "0".

The range mode that the radar has operated in for generating the received PTL can be derived from the bandwidth in the PTL header.

- 720 MHz short range mode
- 365 MHz mid-range mode
- 195 MHz far range mode
- 280 MHz SFMCW-M
- 195 MHz SFMCW-L

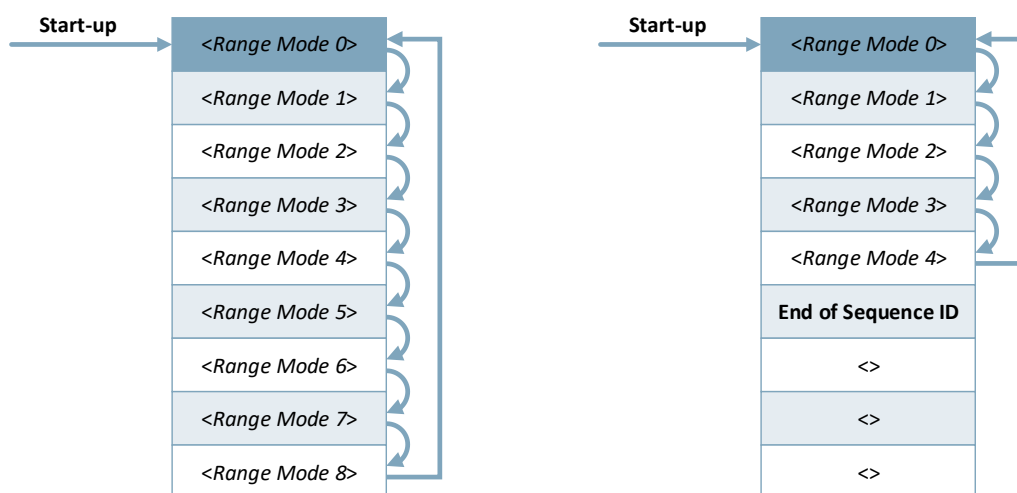


Figure 7: Range mode parameter field

4.8.3.2 COM Control

If the AppPar "RMP Mode" is set to "changeable via COM", the Radar always wakes up in the short range mode after power on. Afterwards, the range mode can be changed by one parameter:

- 4 selects SFMCW-L mode
- 3 selects SFMCW-M mode
- 2 selects far range mode
- 1 selects mid-range mode
- 0 selects short range mode

The selected mode is kept until a new one is given via COM. The change of the range mode is done in the next radar cycle, which is started after the selection is received. If the control shall be cycle precise for the next radar cycle, the COM message shall be sent within the data acquisition phase of the current radar cycle.

Note: The sensor will work as supposed without input data as well.

4.8.4 PRT Mode Control

The PRT mode change can be used for specific applications, e.g., applications that need one PRT only and then derive the benefit of having the right target speed in one shot. However, the host system must ensure that either no speed ambiguity occurs in its ODD or it solves the speed ambiguity by itself.

The PRT mode can be controlled either via COM or via AppPar, selected by the parameter "PRT mode".

The ProWave wakes up after power up in the mode defined by the AppPar "PRT_Mode" or, if COM control is selected, in the alternating mode (enumeration "0"). If a sequence is defined, the sequence starts always at element "0" after power on.

If the PRT mode is defined as COM controlled, then the host can select one of the first 4 values of the enumeration value (see Table 4).

Table 4: PRT Mode control in AppPar

Enum-Value	PRT Mode	Controllability
0	Alternating between PRT low and high for each Radar cycle	AppPar/COM
1	Static high	AppPar/COM
2	Static low	AppPar/COM
3	Sequency	AppPar
4	COM control	AppPar

The sequence for the PRT mode control is defined in a similar way to the range mode selection. The sequence starts at element "0" and ends at the end-of-sequence ID "0xFF". Then it wraps around element "0". If a sequence is selected, the PRT changes once for every radar cycle.

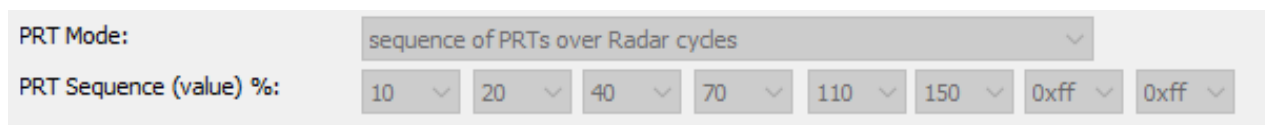


Figure 8: PRT mode parameter field

The PRT cannot be changed to any arbitrary value. There is a lower and an upper limit and the value in between has discrete values. There are more details about this topic in the data description for the AppPar.

4.8.5 Cycle Time and Time Multiplexing

The ProWave offers the option to synchronize the time of the data acquisition windows between several radar sensors in the same vehicle or even other sensors with different detection technology (e.g., LIDAR, camera).

Two parameters can be adjusted to align the data acquisition window of one ProWave with the data acquisition of other sensors.

1. Cycle time
2. Time slot for data acquisition within a radar cycle

Cycle time

This parameter can be adjusted via AppPar from 60ms to 120ms with a scaling of 0.1ms. This adjustment is defined to allow the alignment of cycle times between different sensor technologies (e.g., a Lidar with 66ms and ProWave can both operate with 66ms cycle time).

Time slot for data acquisition

The time slot for data acquisition is defined as the offset to the start of the virtual radar cycle grid. This virtual radar cycle grid is defined by the gPTP time synchronization with the start of the first radar cycle at Jan 1st, 1970, 00:00am UTC (POSIX time), (see Figure 9). This is valid only if the radar time base is synchronized.

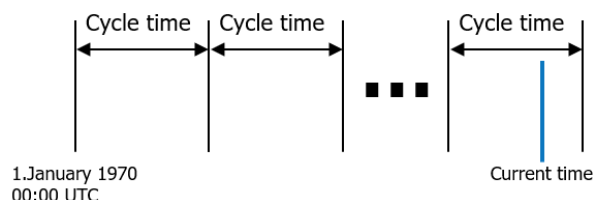


Figure 9: Virtual radar grid

When operating multiple sensors in parallel, time multiplexing helps to avoid mutual interferences from the neighboring sensors. The radar sensors will run synchronized, each of them emitting their RF signal in a different time slot avoiding incoming reflections from neighboring sensors with overlapping FOV.

Figure 10 shows an exemplary setup to apply time multiplexing in a vehicle with multiple radars.

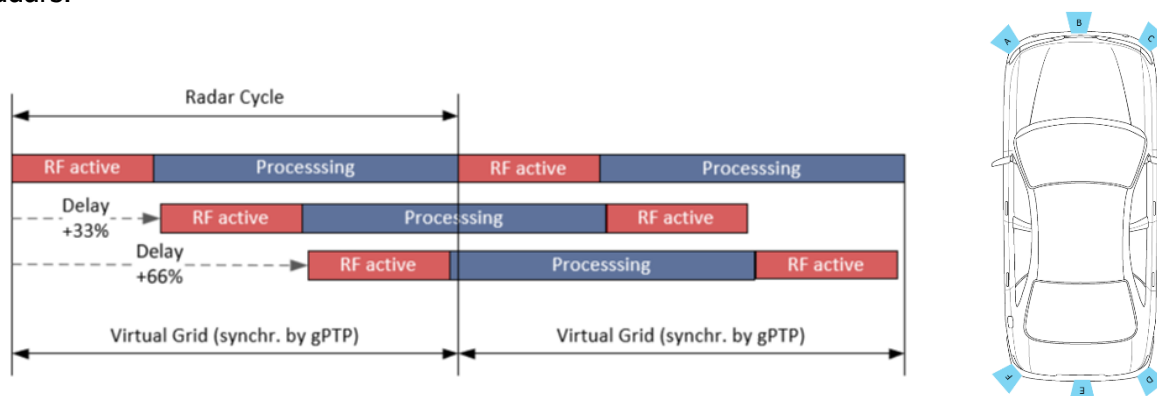


Figure 10: Assignment of time slot for RF transmission and mounting position, example for 6 radars

The "Delay x" to the virtual grid can be adjusted for each sensor in the AppPar. The value has a scaling of 1ms.

Note: Same time slots for diagonal operating sensors should be avoided due to a granted patent in US and CN.

4.8.6 Interference Mitigation Measures

There are several measures for interference mitigation implemented and they can be categorized into two main classes:

1. Modulation and signal processing techniques aimed to counteract radar sources outside of the ego-vehicle.
2. Multiplexing techniques for multiple sensors in one single vehicle.

4.8.6.1 Modulation and Signal Processing

The basic modulation concept ensures that signals from external sources appear as noise in the signal and that no ghost targets are generated. The following state of the art concepts are implemented:

- Nulling of extreme amplitudes in time domain ⇒ It mitigates the effect of external interference from other radar sources in the FoV
- Modulation parameter jitter and phase modulation ⇒ It relaxes the effect of external interference from other radar sources

Interference is typically encountered for a short amount of time. Hence, a distribution of the interference energy to noise is a state-of-the-art approach. A negative effect of this technique is a reduction of SNR, which cannot be avoided but might be acceptable for a limited amount of time. In addition, the grade of performance degradation, even in the presence of interference, is not stable but varies over multiple radar cycles.

This degradation is similar to the impact of snow or heavy rain and must be monitored outside of the sensor. The sensor itself delivers an indicator that he observes some interference but cannot assess the level of performance degradation.

The indicator of jamming is in the point cloud header "JammingStatus" (offset 0x57B) and could have the values "none", "low", "medium", "high", "error".

4.8.6.2 Multiplexing techniques

Since the disturbance caused by multiple sensors in the same vehicle would be permanent, the ProWave offers special measures to avoid the SNR degradation caused by multiple ProWave sensors in one single vehicle. This is the most likely case for L4 applications.

4.8.6.2.1 Time multiplexing

This functionality is already described in paragraph 4.8.5 Cycle Time and Time Multiplexing.

Additionally, to the description in the other paragraph, time multiplexing will avoid interference completely. However, it will also generate latencies between the radar detections of several sensors with overlapping FoVs. Experts of the data fusion in a most likely used perception SW must assess whether this could be afforded or not. It might fit the best for slow vehicle speeds where probably a short-range mode is activated.

4.8.6.2.2 Frequency multiplexing

Frequency multiplexing avoids interference very effectively but there is additional bandwidth needed. Since some of the operation modes don't use the entire 1GHz band this technique can be applied for the mid and far range mode.

Table 5: Sub-band definition

Mode	Short	Mid	Far
Chirp Bandwidth	~900 MHz	~520 MHz	~300 MHz
Subbands	76Ghz 1 77GHz	76Ghz 1 2 77GHz	76Ghz 1 2 3 77GHz

The AppPar allows the user to define whether frequency multiplexing shall be applied for a certain range mode or not. It also allows the user to select the sub-band used.

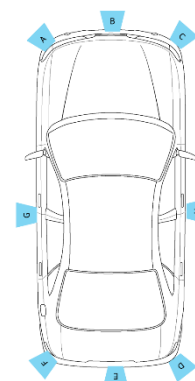
The sensor operates then permanently in this predefined sub-band if the specific range gate mode is selected.

4.8.6.2.3 Chirp slope definition

Since no frequency multiplexing is possible in the short-range mode and there are only two sub-bands for the mid-range mode, another measure is the selection of a different chirp slope. This technique would not avoid interference completely, but it could limit it to a small portion of time. Hence, the impact of interference would be strongly reduced.

Table 6: Application example with 8 radars for frequency multiplexing

RG mode	Mounting Pos	Sub-band	Chirp slope
mid	A, C	1	Up
	B, G, E, H	1	Down
	D, F	2	Up
far	A, C	1	Up
	B, G, E, H	2	Up
	D, F	3	Up
short	A, C, D, F	1	Up
	B, G, E, H	1	Down



4.8.7 Threshold tuning

A short explanation of the signal processing is necessary to understand the threshold tuning. The following sketches show at which step of the signal processing the threshold is used.

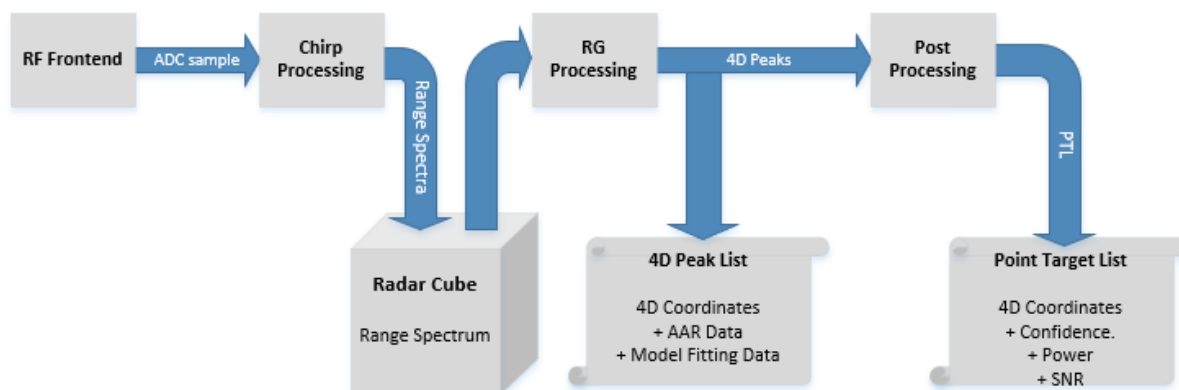


Figure 11: Complete Signal Processing Chain

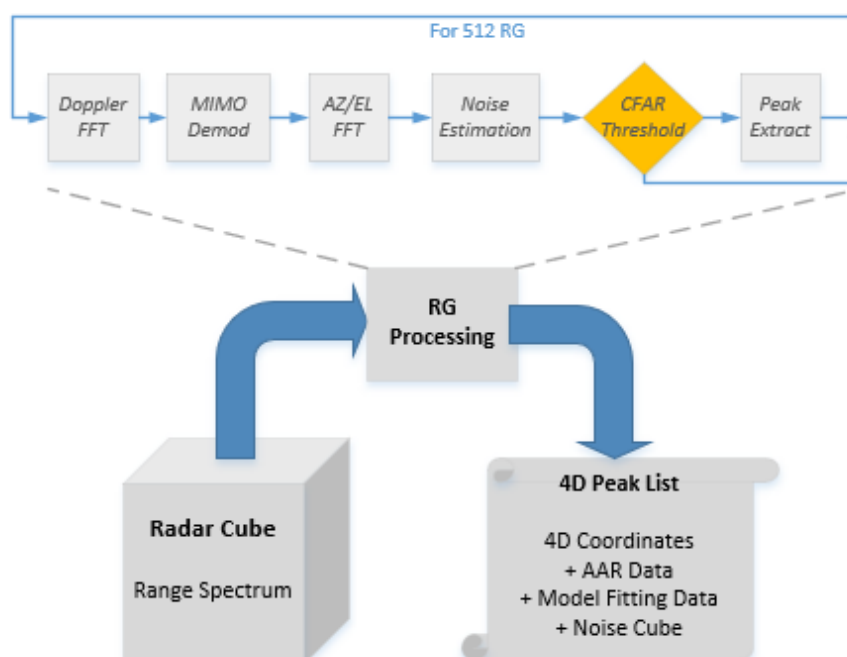


Figure 12: RG Processing

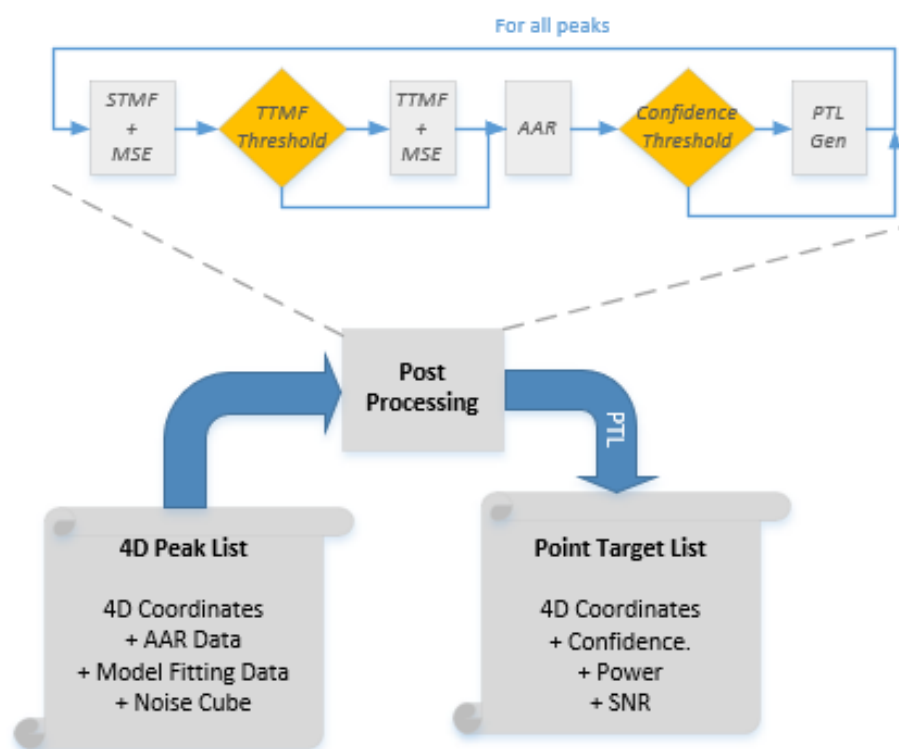


Figure 13: Post Processing

4.8.7.1 CFAR threshold

The CFAR threshold, which is used within the preprocessing stage, distinguishes between valid targets and noise. It is expected that clutter is filtered out at this stage. The standard value for CFAR threshold is 12dB.

The FRR allows certain other values for the CFAR threshold. However, the values are predefined. The current implementation allows the selection of

- 12dB Standard
- 15dB very conservative threshold which reduces the number of targets per cycle
- 9dB more aggressive threshold that increases the number of targets per cycle but generates more false detections.

4.8.7.2 TTMF threshold

TTMF threshold allows the user to influence the amount of times TTMF will be applied to the data. The threshold is used to define whether a TTMF is applied for a certain dimension or not. Nevertheless, the ProWave radar tests first what is the dimension with the biggest deviation to a STMF and applies the TTMF for this dimension only.

The customer cannot change this parameter because it could have significant runtime impact. Therefore, ZF reserves access for this setting. It is possible for ZF to set this value to a customer specific value upon agreement.

The parameter TTMF-threshold allows the use of multiple values. The standard value is 12dB. Lower numbers indicating more and higher number indicating lower number of TTMF-runs per cycle.

4.8.7.3 Confidence filter

The confidence filter throws away targets with low confidence. As explained in paragraph 0, the sensor calculates confidence as weighted sum of all the MSEs per each target point.

The target confidence can have values 0% to 100% but the threshold parameter can be adjusted to 30% or 10% only.

The recommended standard value for this parameter is 30%.

4.9 Cyber Security

The ProWave offers a Cybersecurity Hardware Security Module (HSM) with a safety island according to the EVITA specification, which covers

- Secure communication
- Secure diagnostic
- Secure flash download
- Secure bootup
- Secure locking of the debug port.

4.9.1 Secure Communication

The Secure Communication feature enables the ECU to communicate with other ECUs in the vehicle at runtime.

Secure communication will be handled on UDP packet level by an authenticated signature at the end of each UDP packet. The Secure Onboard Communication (SecOC) is based on the AUTOSAR SecOC. Since the security is just adding an authentication signature (CMAC), the data can be read without knowledge of any key. However, the authentication signature enables the receiver of a UDP packet to check if the transmitter of this message is the right one.

The CMAC is 16 bytes long and AES 128 based. The CMAC will be calculated in the HSM which protects the keys used and provides acceleration to the CMAC operation. The anti-replay value is 8-bytes long (64bit freshness counter).

ZF will share the key and the method to calculate the authentication signature.

4.9.2 Secure Diagnostic

The secure diagnostic feature is designed using the industry standard ISO14229 protocols.

Secure diagnostic is realized based on UDS Security Access, with certificate-based challenge-response scheme. The access control will be implemented based on the Seed and Key generation mechanism. A random number is used for the "Seed" using RSA and a private key in the customers infrastructure for response and a public key in the Radar sensor for verification.

ZF will share the key and the method to calculate the authentication signature.

At a basic level, this is the unlock flow:

1. The tester requests the UDS-Seed from the radar for a particular security level
2. The radar responds with a value to use for UDS-Seed

3. The tester obtains a UDS-Key
4. The tester sends the UDS-Key to the Radar
5. The radar compares the received UDS-Key to the expected value and, if they match, unlocks the security level requested.

4.9.3 Secure Flash Download

The secure flash download is based on top of the secure diagnostic access. The update of the firmware is defined in a way that the host is responsible for

- Checking HW-/SW-compatibility
- Keeping the last version for a requested rollback (since the sensor stores just one image)

If any issue occurs during the firmware download (the image includes the main bootloader as well), the sensor will be still accessible via its fallback bootloader. COM-settings which are defined in AppPar will be also treated by the fallback bootloader. This happens after the next power-up.

Security mechanism of the firmware download are

- HSM-based Signature Verification of Update Packages
- Prevention of Security Downgrading/ Rollback Protection
- Implementation and test hash comparison, signature verification, and downgrade lock with SHA256 and RSA4096.

ZF will share the key and the method to calculate the authentication signature.

4.9.4 Secure Boot Up

The general assessment of the threat, which could be derived from a manipulated image in the flash memory after download, is rated low. The reason for it is that the attacker must have physical access to the ECU and must open the sealed housing. The TARA assumes that a secure flash download is the key feature to protect the ECU against firmware manipulation.

Nevertheless, the secure bootup implementation is prepared, and it is based on a defined process. The elements for it are

- Chain-of-Trust with HW-based Root-of-Trust
- Decryption and Signature Verification of firmware images with Hardware Security Support (HSM)

4.9.5 Secure Locking of the Debug Port

The general assessment of the threat, which could be derived from an open debug port, is rated as low. The reason for it is that the attacker must have physical access to the ECU and must open the sealed housing.

Nevertheless, a deactivation via OTP/fuses or password protection of all Debugging Interfaces (e.g., UART, JTAG/DAP) is foreseen and can be adjusted. Before doing this, the concepts for secure unlock e.g., for warranty analysis – shall be aligned with the individual customer.

5 Electrical and Mechanical Interfaces

The following table identifies the electrical and mechanical specification.

Table 7: Electrical and mechanical specification

Attribute	Value	Comment
Mechanical size	127 x 127 x 42 mm	Sensor body, excluding mounting, connectors and second surface
Weight	<600 g	Weight (C sample) is 588 g
Temperature range	-40°C to +65°C -40°C to +85°C -55°C to +95°C	Operation (without convection) Operation with convection* Storage
Power consumption	18 W (average over one cycle), 31 W (peak)	
Operating voltage	+8.5 V to +16 V DC	
Lifetime	8,000 operational hours	Corresponds up to 300,000km with assumed speed profile.
Electronic assembly process	Lead free (ELV compliance)	
Vibration resistance	According to LV124	
Protection class	IP6K9K	Dust, high pressure, water spray
Chemical resistance	ISO 16750-5.	Diesel & petrol fuel, battery acid hydraulic, and brake fluid, engine and transmission lubricant, anti-freeze, sealant and its removal agent, coke, sweat and glass cleaner.
EMC component emissions:	CISPR 25 Ed 3 (2008) Class 3	
EMC component susceptibility:	ISO11452_2 Ed 2 (2004) Class 4 (RI) ISO11452_4 Ed 4 (2011) Level 3 (BCI)	

*To achieve 85°C maximum temperature, vehicle integration must provide a 1m/s airflow around the radar. Further recommendations can be found in "FRGen21 C Sample Mechanical Integration Guideline".

Figure 14 depicts the sensor dimensions and the position of the power and data connector.

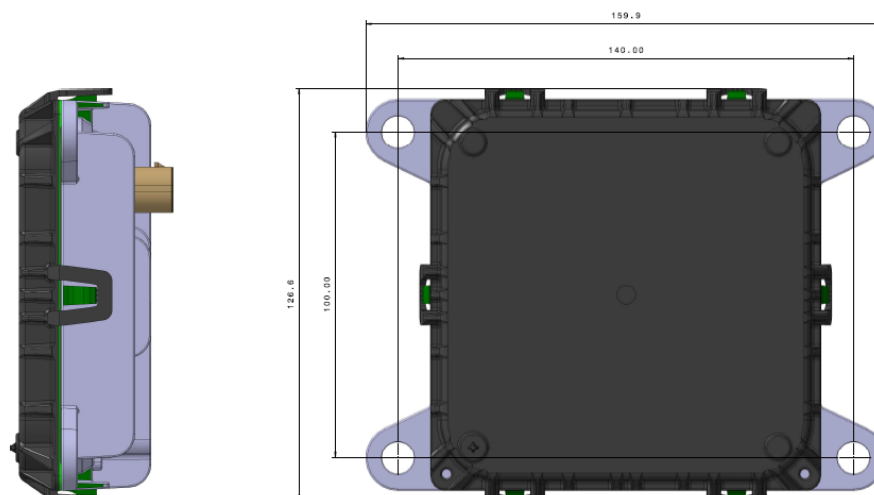


Figure 14: Radar sensor dimensions (C sample)

5.1 Sensor mounting

The sensor can be mounted to the vehicle directly. Both, electrical isolation and electrical contact is allowed. Further mounting recommendation is given in section 6.

5.2 Power Supply

The power supply, shown in Figure 15, is manufactured by Hirschmann. The drawing shows the pin count of the connector, which corresponds to the pin number in Table 8.

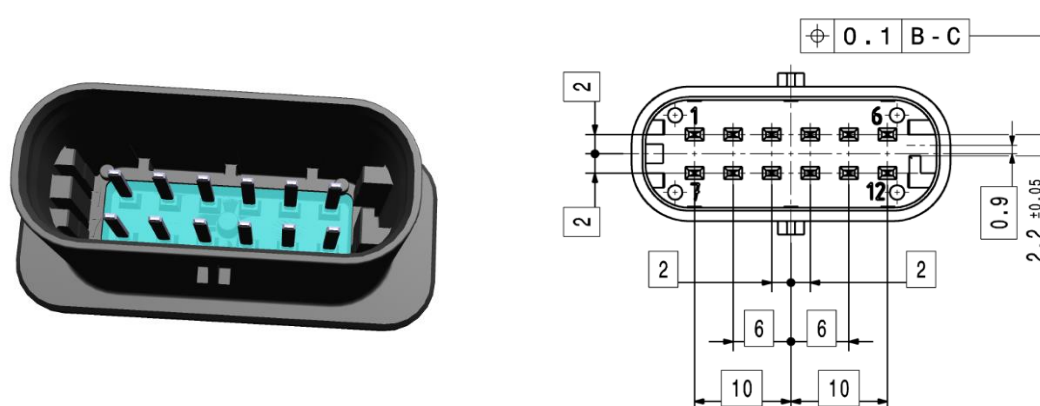


Figure 15: Picture and drawing of the power connector from Hirschmann on sensor side

The pin specification is given in Table 8 with the referred pin number, function, and nominal values for voltages and current.

Table 8: Pin assignment of power connector

Pin	Pin function	Pin type	Nominal voltage	Nominal current	Wiring requirement
1	Ignition signal for wake	IN	13.5 V	10 mA	0.5 mm ² Color: black
2	NC	-	-	-	-
3	NC	-	-	-	-
4	NC	-	-	-	-
5	NC	-	-	-	-
6	NC	-	-	-	-
7	NC	-	-	-	-
8	Sensor V _{BAT}	IN	13.5 V	1500 mA (0 to 3.5 A)	1.0 mm ² Color: black
9	NC	-	-	-	-
10	Ground to vehicle	IN	0 V	1500 mA (0 to 3.5 A)	1.0 mm ² Color: black
11	NC	-	-	-	-
12	NC	-	-	-	-

The corresponding harness for the power connector in the length of two meter will be part of the standard delivery of the B-sample. Different length can be ordered from ZF-XA. Please contact your corresponding ZF sales representative.

For series, the cables (pigtailes) including cable side connectors shall be ordered directly from Hirschmann. The Hirschmann part number of the cable connector is HA 806-344-541.

5.3 Data Connector

The sensor applies the standard Automotive Ethernet 100BASE-T1 (BroadR-Reach®) IEEE 802.3bw-2018 for the data connection and has two Ethernet connections (A and B). The Ethernet connection A is used for the standard communication, where the sensor delivers the point cloud. The Ethernet connection B is not offered on series HW. It is only used during development as a debugging connection.

The picture and the drawing in figure 16 show the connector on sensor side.

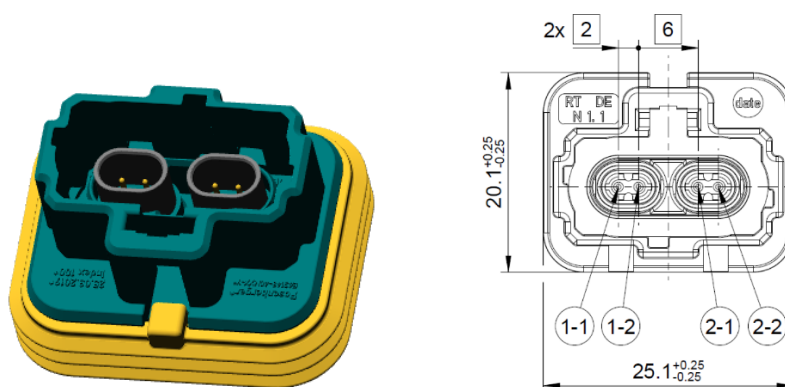


Figure 16: Picture and drawing of the data connector from Rosenberger on sensor side

The pin specification is given in table 9 with the referred pin number and the corresponding signal.

Table 9: Pin assignment of the data connector

Pin	Pin function	Pin type	Nominal voltage	Nominal current	Wiring requirement
1-1	Ethernet channel A – negative	IN/OUT	2.2 V (differential)	100 mA	0.5 mm ² Color: green 72 turns per m
1-2	Ethernet channel A – positive	IN/OUT			0.5 mm ² Color: white/green 72 turns per m
2-1	Ethernet channel B – negative	IN/OUT	2.2 V (differential)	100 mA	0.5 mm ² Color: blue 72 turns per m
2-2	Ethernet channel B – positive	IN/OUT			0.5 mm ² Color: white/blue 72 turns per m

The corresponding harness for the data connector in the length of three meters will be a part of the standard delivery of the B-sample. Different length can be ordered from ZF-XA. Please contact your corresponding ZF sales representative.

For series, the cables (pigtailed) including cable side connectors shall be ordered directly from Rosenberger. The Rosenberger part number of the cable connector is E6K14K-1CAZ5-Y.

5.4 Secondary Surface

The sensor is not designed to withstand strong mechanical shocks, e.g., stone chipping. Therefore, if the installation height is very low (stone chipping is very likely), it is recommended to use a secondary surface in front of the sensor.

Since the design of such a surface is very strongly connected to the vehicle design itself, ZF does not offer such a secondary surface. Nevertheless, ZF can assist the design of such secondary surface with the following support options

- Delivering of an RF installation guideline, which contains some recommendations on how ProWave shall be installed from a radar perspective
- Characterizing material according to its RF attributes, by measuring a material probe in the size of about min. 10x10cm
- RF-Simulation of complete mounting configuration including secondary surface.

6 Vehicle integration

For vehicle integration, ZF supports the customer with a CAD file of the ProWave sensor. This CAD file includes (dependent on HW state even more information might be included)

- the sensor
- the radar phase center
- the keep-out zone

6.1 RF Integration and Keep-Out Zone

There is additional documentation available, which describes the RF integration aspects in the vehicle. Please refer to

- FRGen21 RF Integration PRO_SPC_2100.pdf RF integration

6.2 Sensor Mounting

There is additional documentation available that describes mechanical and electrical integration in the vehicle. Please refer to

- FRGen21 C Sample Mechanical Integration Guideline.pdf Mechanical integration

6.3 Intrinsic and Extrinsic Calibration

6.3.1 Intrinsic calibration

Every ProWave sensor obtains an initial phase calibration at the production line from ZF. This phase calibration ensures proper performance over the full temperature range and over the entire product life. An online phase calibration is not implemented.

6.3.2 Extrinsic calibration

The ProWave guarantees a maximum boresight error of 0.4° by mechanical tolerance calculation. However, no misalignment correction or even misalignment compensation is implemented in the ProWave. Since the sensor outputs just polar coordinates of the point targets in its FoV, it does not consider the mounting position or any misalignment. All angles are referred to boresight vector of the sensor itself.

6.4 Conformance with Regulatory Frequency Requirements

The sensor complies with the following RF emission standards and characteristics:

Table 10: Frequency homologation

Range	Limits Europe	Limits USA	Measurement Data (Typical)
Frequency	76-77 GHz	76-77 GHz	76-77 GHz
Main Standard	ETSI EN 301 091	FCC Part 95, Subpart M	
Bandwidth	1 GHz	1 GHz	0,23 GHz (99% OBW) 0,43 GHz (99% OBW) 0,84 GHz (99% OBW)
EIRP peak	55 dBm	55 dBm	44 dBm
EIRP mean (RMS)	50 dBm	50 dBm	33 dBm

In addition, the radar sensor complies with the two following EMC standards

- ETSI EN 301 489, including ISO 7637 ("Electro Magnetic Compatibility (EMC) standard for radio equipment and services")
- ECE Reg. 10 ("Uniform provisions concerning the approval of vehicles with regard to electromagnetic compatibility")

6.5 Human Health

The sensor complies with the following legal limits for the limitation of exposure of the general public to electromagnetic fields as shown:

Table 11: Human health limits

Range	Limits Europa	Limits USA	Measurement Data (Typical)
Main Standard	DIN EN IEC 62311 VDE 0848-311:2020-12 Bewertung von elektrischen und elektronischen Einrichtungen in Bezug auf Begrenzungen der Exposition von Personen in elektromagnetischen Feldern (0 Hz bis 300 GHz) 1999/519/EC: Council Recommendation of 12 July 1999 on the limitation of exposure of the general public to electromagnetic fields (0 Hz to 300 GHz)	FCC §1.1310 Radiofrequency radiation exposure limits. FCC § 2.1091 Radiofrequency radiation exposure evaluation: mobile devices.	See EU and FCC test report
Power density limit	$\leq 10 \text{ W/m}^2 = 1\text{mW/cm}^2$ at a distance $\geq 20\text{cm}^*$	$\leq 10 \text{ W/m}^2 = 1\text{mW/cm}^2$ at a distance $\geq 20\text{cm}^*$	$0,3 \text{ mW/cm}^2$ at a distance $\geq 20\text{cm}$
Class	Basic Restriction General Public	Limits for General Population/Uncontrolled Exposure §1.1310(e)(1)-Table 1(ii)	
Warning sign	No warning sign (radiation hazard) required		

* EU and FCC limits correspond to the recommendations of the "INTERNATIONAL COMMISSION ON NON-IONIZING RADIATION PROTECTION, ICNIRP PUBLICATION – 2020, ICNIRP GUIDELINES FOR LIMITING EXPOSURE TO ELECTROMAGNETIC FIELDS (100 KHZ TO 300 GHz), Table 5: Reference levels for exposure, averaged over 30 min and the whole body, to electromagnetic fields from 100 kHz to 300 GHz (unperturbed rms values)", published in: HEALTH PHYS 118(5): 483–524; 2020

- The sensor is classified as a mobile device and as such, radiation power density is evaluated at a distance of $\geq 20 \text{ cm}$.
- The sensor meets the limits for General Public (EU/ICNIRP) and General Population/Uncontrolled Exposure as per FCC §1.1310(e)(1)-Table 1(ii) and as such does not require a warning sign (radiation hazard)
- The usage of the higher "Limits for Occupational/Controlled Exposure" as per §1.1310(e)(1)-Table 1(i) which are $\leq 50 \text{ W/m}^2 = 5\text{mW/cm}^2$ is not required.

7 Operation

To ensure the correct function of the sensor, the following input and output signals must be set and interpreted properly.

7.1 Wake-Up and Shutdown Strategy

According to the pinning, described in the previous chapters, the sensor can be connected to the vehicle and operated with different options. The ProWave can operate in the following modes:

- KL15 mode (switched power supply)
- KL30 mode (permanent power supply plus additional wakeup).

The ProWave radar does not support wakeup via COM. Hence, ProWave offers wakeup just via a discrete line. The difference between KL15 and KL30 operation is the size of the required switch in the power supply of the vehicle. During the KL15 operation, this switch shall be able to withstand a couple of Ampere peak current (for about 100 μ s) during power on. The wakeup-switch of a KL30 system has to handle a couple mA to switch the radar on. However, KL30 requires at least up to 100 μ A quiescent current.

7.1.1 KL15-Operation

The KL15 operation is quite simple. The radar will be switched on and off just by the main power supply. The wakeup signal shall be connected to the Vbat either from harness side or from the central power supply (see Figure 17).

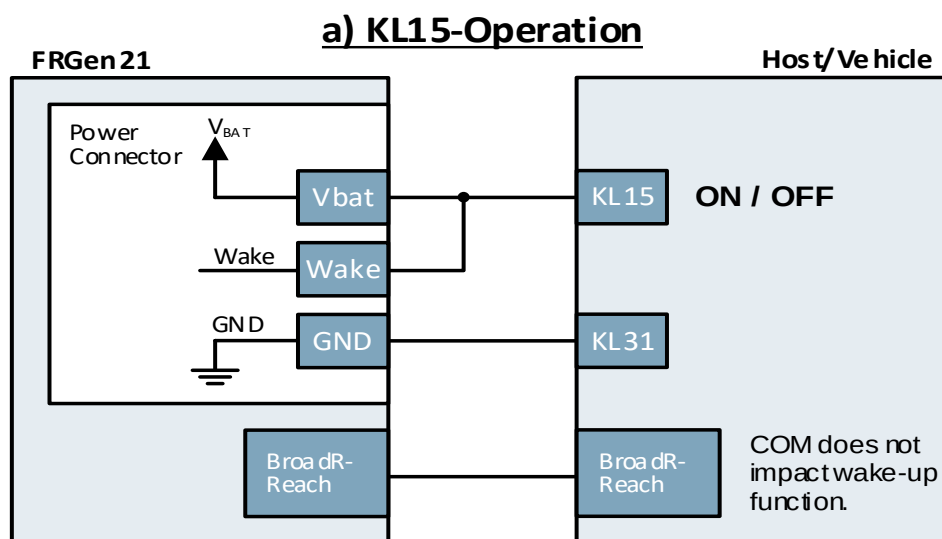


Figure 17: KL15 Operation

7.1.2 KL30-Operation

The KL30 operation is depicted in Figure 18. The radar is supposed to be powered with permanent power on "Vbat" and will be switched on/off by the discrete wakeup line on connector pin "Wake". The behavior of the SW might be such that it slightly postpones the

shutdown after it recognizes the low status on "Wake". Nevertheless, this delay is guaranteed to be lower than 100ms.

Limits:

Table 12: KL15 limits

Limit	Value
KL15 on-threshold	See HW Release Note Vehicle Interface
KL15 off-threshold	See HW Release Note Vehicle Interface
SW KL15 logic on	> 6V
SW KL15 logic off	< 4V
Delay KL15 to booting	<20ms
Delay booting to SW is running	<5sec
Switch off delay	<100ms after shutdown command

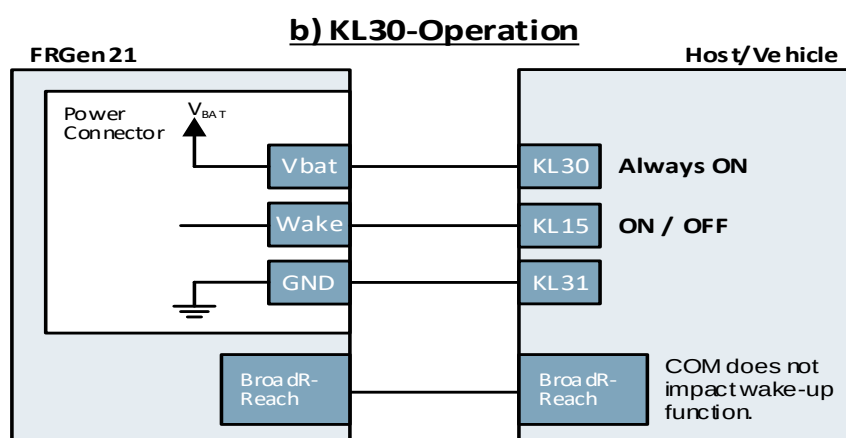


Figure 18: KL30 Operation

8 Data Interfaces

8.1 General

All communication of the sensor is based on UDP network protocol, except for the diagnostic services using UDS protocol which uses a TCP/IP protocol as basis.

8.2 Input Interface

8.2.1 Description

There is one data structure defined for the input data which is defined as a C-header file as well as an Excel file with additional information, e.g., word sizes, scaling factor etc. The files are called

- C-header: RxSignalHandler_Cfg_Cmp.h
- Description: FRGen21_InputData[_Rx.x].xlsx

The name of the struct in the C-header file is called "CommsInputSignals".

8.2.2 COM Structure

The input data are packed in one single UDP packet, which contains 9 data bytes. The input data shall be sent at least one time per radar cycle since there is a self-diagnostic monitor in the sensor that watches this frequent reception as keep alive message for the host connection.

8.3 Output Interface

The ProWave essentially outputs two data packets. The detection data is collected in the so-called Point Target List (PTL) and additional status data is provided at a different IP port.

8.3.1 Point Target List

8.3.1.1 PTL Description

The point target list represents the measurement result of one radar cycle. Each PTL consists of a header with certain system parameters and an array of multiple point targets. The structure of the PTL is defined and described in

- C-header: ZF_xxRadarPointTargetInterface.h
- Description: FRGen21_PTL[_Rx.x_xx].xlsx

All data in the PTL are organized in little endian format.

The sensor outputs some primary parameters for each target point, such as the frequencies, corresponding to each of the 4 dimensions: range, velocity, azimuth, and elevation. The header of this list also contains some context data about the modulation that is used by the physical value derivation routines to do the transformation from frequencies to SI values. These little C++ routines are delivered with each SW package and are supposed to run on the host ECU.

The format of the point target descriptors is defined in this special way to have the option to express ambiguities in the angle dimensions as one single value.

Some special elements of the PTL-header are described in more detail in the following table.

Table 13: Header of the PTL

Name	Part of	Description
uiNbOfTargets	Context Data	Number of valid point targets. The maximum which is possible from the COM structure is 3,350 points but the radar will deliver, depending on threshold setting, about 2,000 points.
TimeStampMeasure Start/End	Context Data	It indicates the time of the gPTP synchronized time base at start of chirping and end of chirping.
eFullTargetList	Context Data	It indicates whether the internal signal processing has been successfully finished or stopped by the radar cycle timeout.
MaxDetectionRange	Context Data	The ProWave processes range gates sequentially. This parameter indicates the maximum distance the radar has processed detection data for. It can be used to check whether there are any potential missed detections due to internal processing limitations and, if so, beyond which distance are these missed detections expected.
Data Validity	Context Data	It indicates the validity of the whole PTL. The PTL is set to invalid in case of an internal error.
Dimensions AntennaElementGain AntennaSpacing	DynamicConfig Params	These substructures contain data that is used by the physical value derivation routines.
PRT	DynamicConfig Params	This value in this substructure is supposed to be used to solve the speed ambiguity (see section 4.1 and 4.5).
JammingStatus	Blockage Detection	This parameter indicates the grade of jamming observed by the sensor itself. But the degradation of the detection performance shall be derived by a blockage algorithm running on the host ECU.

Each point target is described by several parameters. Details can be obtained out of the description file "FRGen21_PTL[_Rx.x_xx].xlsx". The following table gives a brief description of these parameters.

Table 14: Parameter of one-point target

Name	Description
Frequency Range Velocity Azimuth angle Elevation angle	These are the detected frequencies for the target point in each of the 4 dimensions: range, velocity, azimuth, and elevation. They can be transformed to physical values (SI units) by using the C++ routines defined in "xxPhysicalValueDerivation.cpp"
Mean-squared fitting error	It indicates the fitting error to the assumed target model in each dimension. See description in 4.7
Model type	It indicates which target model has been used. The sensor uses either a single target model or a two-target model.
SNR	Signal to noise ratio of the single point target.
Detection confidence	Weighted sum of all the MSE values. It indicates how well did the model fitting and the angular estimation work for the target point.
Peak ID	It indicates the original peak where this point target is derived from.
Power	Interpolated power for the target point. The interpolated power is not just the power of the closest FFT bin but the value between two FFT bins in each dimension.

8.3.1.2 Derived Parameters

The primary parameters provided for range, velocity, azimuth, and elevation angle in the PTL are frequency-based representations and must be converted to physical values as a post-processing step from the customer. Such conversion can be performed by using the C++ routines provided by ZF ("B2PhysicalValueDerivation.cpp"). The transformation from frequency domain to physical values is done based on a sensor-centered spherical coordinate system.

Figure 19 depicts the orientation of the coordinate system to the ProWave sensor. The considered spherical coordinate system has the sensor centered in the y-z plane. Here, ϕ denotes the azimuth angle, θ the polar angle and r the range. The elevation angle is $\pi/2 - \theta$.

To calculate the physical values based on the vehicle coordinate system, the customer must compute a misalignment compensation.

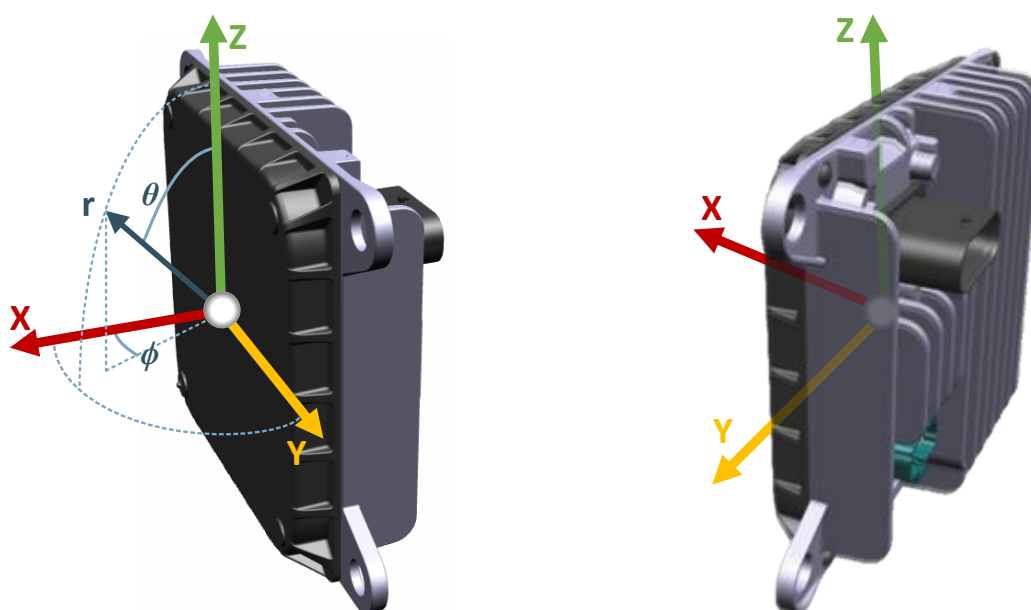


Figure 19: Orientation of the sensor coordinate system

To calculate the physical values based on the vehicle coordinate system the customer has to compute a misalignment compensation.

8.3.1.3 PTL COM Structure

The PTL is defined as a fixed structure which is filled up partly by every cycle. The PTL indicates the number of valid point targets (`uiNbOfTargets`) in the header and has a size of 65,012 bytes (XA PTL). Due to some AUTOSAR limitations, the whole structure is split on the application layer into 7 UDP packets. Six of them are containing 10,352 bytes and the 7th packet contains 2,900 bytes of the PTL.

As the ETH stack cannot send 10,352 bytes within 1 package, the ETH stack of the AUTOSAR layer fragments the payload into several messages and the socket interface of the customer's receiver defragments those packages automatically into the 10,352 bytes. The customer then manually combines the six 10,352-byte packages with the 7th packet (2,900 bytes) to obtain the full PTL.

The effective transfer speed on the 100BaseT is up to 40-45Mbps. The transfer of the whole PTL takes about 13-15ms.

8.3.2 Status Data/Health Monitoring

8.3.2.1 Description

The status data are used to indicate internal status information of the sensor. The status data are supposed to be just logged by the customer side. Some of the status data are described in Table 15. The remaining status data are not described to the customer but give ZF the chance for a quick analysis in case of abnormal behavior of the radar.

The status data are defined and described in the following file

- C-header: A53StateData.h

Table 15: ProWave Status Data

Name	Description
System State Manager	It indicates the system state the radar is currently operating on.
Cycle Count	It can be used to observe whether the radar has been internally stopped or it operated normally.
Several Temperatures	It can be used to assess whether the radar gets abnormally hot. The radar operates internally about 40-45°C above the ambient temperature.
Battery voltage	The radar measures the battery voltages on its internal entry stage of the power supply.
Received speed	It can be used to confirm the reception of the input parameter "VehicleSpeed".
Version	It describes the version of the status data structure.

8.3.2.2 COM Structure

The status data are packed in one single UDP packet, which contains 1,400 data bytes. This size ensures that it can be handled as a single IP packet.

9 Software Update

9.1 General

A Windows-based tool (ZF-XA-flash tool) is available to flash each sensor with the correct SW. The ZF-XA-flash tool checks what HW is connected and whether the binary that is selected to download fits to this HW.

However, there is one important remark to this tool. This flash tool will be only supported for development phase. For series, ZF is assuming that flashing will be integrated in a customer SW environment.

ZF is not providing a flash tool qualified for series usage!

9.2 Flashing during Development

Using this development tool, the tool itself shall be installed on a Windows PC and a license key shall be requested from ZF. This is described in the Flash-Instruction_Verx.x.x.x.pdf. The executable of the Windows flash tool can be shared along with the release notes to a SW release distribution.

The flash instruction describes the necessary hardware and software to flash the unit.

The flashing tool supports both, the download of new SW and download of application parameters.

9.3 Flashing during Series

ZF has concluded, after receiving feedback from several customers, that it is the preferred choice for the customers to integrate the flashing into their own SW environment. Hence, the flashing concept for series is slightly different and a Windows based tool with a human user interface is not the solution.

ZF can provide the sensor firmware as a SW container *.frgen21 including binary file (image). In addition, ZF can deliver a description of a sequence of UDS calls to perform both the

- SW binary download and
- AppPar download

Flow diagrams are shown in Figures 20+21. Further details can be provided on request.

Parameter changes will be handled like SW flashing, but the sensor will perform more internal checks.

There will be a roll-back functionality for COM parameters to ensure that an IP-address change will not lead to a loss of the communication in general (e.g., by losing the network class if the sensor switches back to a default address).

The sensor checks for all other parameters range or individual enumeration values to avoid implausible settings in the AppPar.

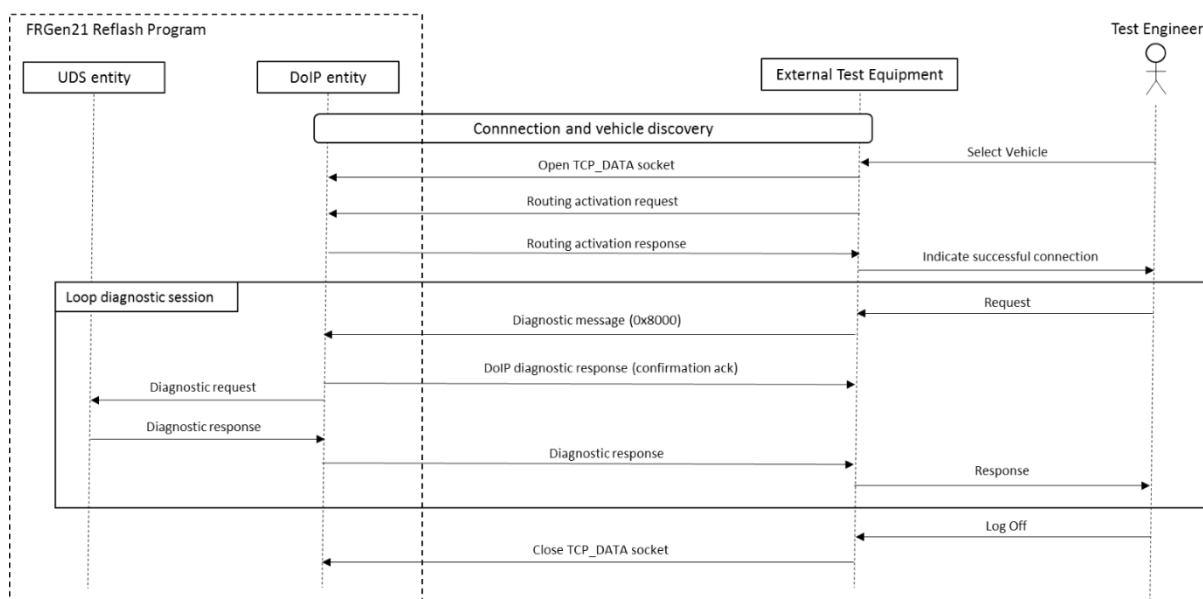


Figure 20: Flow diagram (general overview) of the reflash sequence.

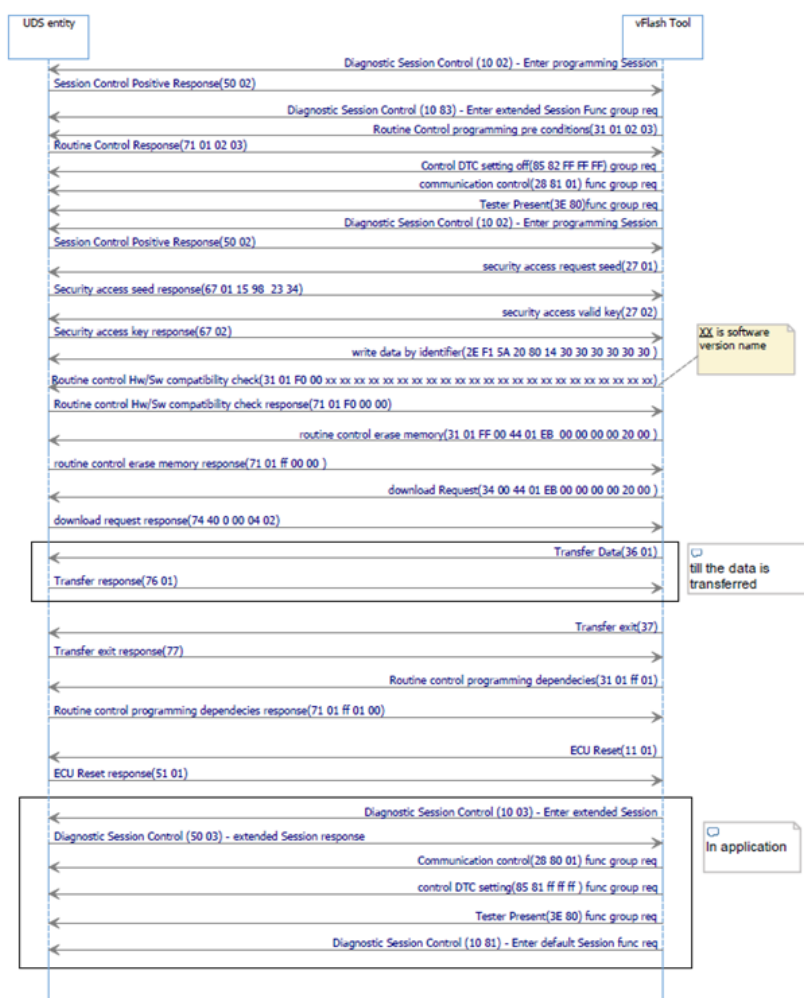


Figure 21: Reflash sequence.

10 Ordering Information

10.1 Scope of Delivery

Series products are supposed to be shipped to the customer without

- Bracket and clips
- Secondary surface
- Harness for power
- Harness for COM
- Executable of Physical Value Derivation (C-Code template available)
- Flash tool

10.2 Ordering of Add-On's

Table 16: Ordering information

Option	Supplier	Ordering No.
IMU	ZF Friedrichshafen AG	
Additional Ethernet (Port B)	ZF Friedrichshafen AG	
Mating connector for power supply	Hirschmann	HA 806-344-541
Mating connector for data connection	Rosenberger	E6K14K-1CAZ5-Y



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ISO 9001

ISO 14001

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